LEVEL 15 · PART VII

Disaster Recovery

15

Goal: Design, document and rehearse recovery so that failures are survivable and recovery times are known, not guessed.

RECOMMENDED TIME
2-3 weeks

TOPICS
10

PRACTICE / CAPSTONE
Practice failure scenarios: database failure, server failure, pod failure, node failure and region failure, with timed recovery and a written DR plan.

Topics

#	Topic	What you will learn
1	Backups **p.2985**	3-2-1 rule, full/incremental, logical vs physical, encryption, retention
2	Replication **p.2997**	sync/async, cross-AZ/region, replication is not backup
3	Snapshots **p.3007**	EBS/RDS snapshots, consistency, lifecycle, cross-region copy
4	Restore Procedures **p.3019**	documented runbooks, order of restore, validation after restore
5	Failover **p.3033**	automatic vs manual, DNS/LB failover, split-brain, failback
6	RPO - Recovery Point Objective **p.3047**	data-loss tolerance, deriving backup frequency, cost of lower RPO
7	RTO - Recovery Time Objective **p.3059**	downtime tolerance, DR tiers (backup/restore, pilot light, warm standby, active-active)
8	Off-site Backups **p.3073**	cross-region/cross-account, immutability, object lock, ransomware protection
9	Testing Backup Restoration **p.3089**	automated restore tests, checksums, restore timing, evidence
10	Failure Scenario Drills **p.3103**	database, server, pod, node, region failure drills and game days
★	Level capstone project **p.3121**	Practice failure scenarios: database failure, server failure, pod failure, node failure and region failure, with timed recovery and a written DR plan.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

1 Backups

2 Replication

3 Snapshots

4 Restore Procedures

5 Failover

6 RPO - Recovery Point Objective

7 RTO - Recovery Time Objective

8 Off-site Backups

9 Testing Backup Restoration

10 Failure Scenario Drills

★ Capstone